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Problem 1: Singly Linked List - In-Place Reversal and Pointer Tracing Given a singly linked list head → [10] → [25] → [40] → [15] → NULL.

- Task 1: Write a function / algorithm in pseudocode to reverse the simply linked list including three pointer (prev, curr, next) after each.
- Task 2: Pointer trace table.

~~Problem 2: Circular linked list - Insert at Head Given circ~~

Task 1: Funcion Reverse linked list (head)

prev = NULL

curr = head

next = NULL

While curr != NULL:

next = curr.next

curr.next = prev

prev = curr

curr = next

head = prev

Return head

Task 2: Initialization: prev = NULL, curr = node(10), next = NULL.

Iteration / Step	prev	curr	next	Linked list Structure. State
Initial	NULL	[10]	NULL	[10] → [25] → [40] → NULL
Interation 1	[10]	[25]	[25]	NULL ← [10] | [25] → [40] → [15] → [NULL]
Interation 2	[25]	[40]	[40]	NULL ← [10] ← [25] | [40] → [15] → NULL.

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Iteration 3 : [40] [15] [15] NULL ← [10] ← [25] ← [40]

← NULL.

Iteration 4 : [15] NULL NULL NULL ← [10] ← [25] ← [40] ← [15]

Reversed list: head → [15] → [40] → [25] → [10] → NULL

* Problem 2 Circular Linked List - Insert at Head : 20

• Task 1 : Pseudocode for InsertHead . 0,75

Function InsertHead (head, newValue) :

newNode = CreateNode (newValue) :

If head == NULL:

newNode . next = newNode .

Return newNode

tail = head .

While (tail next != head) :

tail = tail . next .

newNode

~~head~~ . next != head ? newNode . next = head .

tail . next = newNode

head = newNode

Return head .

1,25 • Task 2 : Execution for Inserting

1. Create node : newNode = [5]

2 . Find tail : Traverse [10] → [20] → [30] . [30] next == head, stop at tail = [30]

3 Link new node newNode .

~~- Singly linked list~~ K

newNode.next = head ([5].next → [10]).

4. Link tail: tail.next = newNode ([30].next → [5]).

5. Update head: head = newNode (head points to [5].

head → [5] → [10] → [20] → [30] → (points back to [5])

Problem 1: 2.5

Problem 2: 2.0

Problem 3: 0.0

Problem 4: 0.0

4.5

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